Classical Schur–Weyl Duality with Explicit Matrix Elements

Notes

Abstract

We give a self-contained account of classical Schur–Weyl duality for $\mathfrak{gl}_N$ and S_n, emphasising the explicit construction of matrix elements for both actions. The Robinson–Schensted–Knuth correspondence emerges as the combinatorial incarnation of the decomposition: the pair (P, Q) produced by RSK literally *is* the Schur–Weyl decomposition, giving coordinates in the $\mathfrak{gl}_N$ and S_n factors simultaneously.

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Part I
Classical Schur–Weyl Duality

1 The two commuting actions

Fix a positive integer N and let $V = \mathbb{C}^N$ with standard basis $e_1, \dots, e_N$. For any positive integer n, the tensor power $V^{\otimes n}$ has dimension N^n with basis

$$e_{\mathbf{i}} \;=\; e_{i_1} \otimes e_{i_2} \otimes \cdots \otimes e_{i_n}, \qquad \mathbf{i} = (i_1, \dots, i_n) \in \{1, \dots, N\}^n.$$

We call $\mathbf{i}$ a *word* of length n in the alphabet $\{1, \dots, N\}$.

Two groups act naturally on $V^{\otimes n}$.

1.1 The GL_N action

The general linear group $\mathrm{GL}_N(\mathbb{C})$ acts *diagonally*:

$$g \cdot (e_{i_1} \otimes \cdots \otimes e_{i_n}) = g\, e_{i_1} \otimes \cdots \otimes g\, e_{i_n}.$$

In matrix elements, if $g = (g_{ab})$ then

$$g \cdot e_{\mathbf{i}} = \sum_{\mathbf{j} \in \{1,\dots,N\}^n} \Big(\prod_{k=1}^{n} g_{j_k, i_k} \Big)\, e_{\mathbf{j}}.$$

The corresponding Lie algebra action of $\mathfrak{gl}_N$ is given by the matrix units E_{ab} $(1 \leqslant a, b \leqslant N)$ acting as

$$E_{ab} \cdot e_{\mathbf{i}} = \sum_{k=1}^{n} \delta_{b, i_k}\, e_{i_1} \otimes \cdots \otimes e_{i_{k-1}} \otimes e_a \otimes e_{i_{k+1}} \otimes \cdots \otimes e_{i_n}. \tag{1}$$

That is, E_{ab} scans each tensor factor and replaces e_b by e_a wherever it finds it.

1.2 The S_n action

The symmetric group S_n acts by *permuting tensor positions*:

$$\sigma \cdot (e_{i_1} \otimes \cdots \otimes e_{i_n}) = e_{i_{\sigma^{-1}(1)}} \otimes \cdots \otimes e_{i_{\sigma^{-1}(n)}}. \tag{2}$$

Equivalently, σ sends the word $\mathbf{i} = (i_1, \dots, i_n)$ to the word $\sigma \cdot \mathbf{i} = (i_{\sigma^{-1}(1)}, \dots, i_{\sigma^{-1}(n)})$. Since S_n is generated by adjacent transpositions $s_k = (k,\ k{+}1)$ for $k = 1, \dots, n{-}1$, it suffices to specify:

$$s_k \cdot e_{\mathbf{i}} = e_{i_1} \otimes \cdots \otimes e_{i_{k+1}} \otimes e_{i_k} \otimes \cdots \otimes e_{i_n},$$

which simply swaps positions k and $k{+}1$.

1.3 Commutativity

Proposition 1.1. *The actions of* GL_N *and* S_n *on* $V^{\otimes n}$ *commute.*

Proof. For any $g \in \mathrm{GL}_N$ and $\sigma \in S_n$,

$$\begin{aligned} g \cdot (\sigma \cdot e_{\mathbf{i}}) &= g \cdot (e_{i_{\sigma^{-1}(1)}} \otimes \cdots \otimes e_{i_{\sigma^{-1}(n)}}) \\ &= g\, e_{i_{\sigma^{-1}(1)}} \otimes \cdots \otimes g\, e_{i_{\sigma^{-1}(n)}}, \end{aligned}$$

while

$$\begin{aligned} \sigma \cdot (g \cdot e_{\mathbf{i}}) &= \sigma \cdot (g\, e_{i_1} \otimes \cdots \otimes g\, e_{i_n}) \\ &= g\, e_{i_{\sigma^{-1}(1)}} \otimes \cdots \otimes g\, e_{i_{\sigma^{-1}(n)}}. \end{aligned}$$

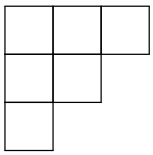
The key: g acts within each tensor factor (it doesn't see position), while σ rearranges factors (it doesn't see which basis vector is in each slot). □

2 The decomposition

2.1 Partitions and Young diagrams

A *partition* λ of n, written $\lambda \vdash n$, is a weakly decreasing sequence $\lambda = (\lambda_1 \geqslant \lambda_2 \geqslant \cdots \geqslant \lambda_\ell > 0)$ with $|\lambda| = \lambda_1 + \cdots + \lambda_\ell = n$. We write $\ell(\lambda) = \ell$ for the number of parts. The *Young diagram* of λ consists of λ_i left-justified boxes in row i (using English convention, with row 1 at the top).

Example 2.1. The partition $\lambda = (3, 2, 1) \vdash 6$ has Young diagram



2.2 Standard and semistandard tableaux

Definition 2.2. A *standard Young tableau* (SYT) of shape $\lambda \vdash n$ is a filling of the Young diagram of λ with the numbers $1, 2, \ldots, n$, each used exactly once, such that entries increase along each row (left to right) and down each column (top to bottom).

Definition 2.3. A *semistandard Young tableau* (SSYT) of shape λ with entries in $\{1, \ldots, N\}$ is a filling of the Young diagram of λ such that entries weakly increase along each row and strictly increase down each column.

We write $\mathrm{SYT}(\lambda)$ for the set of standard tableaux of shape λ, and $\mathrm{SSYT}(\lambda, N)$ for the set of semistandard tableaux of shape λ with entries in $\{1, \ldots, N\}$.

Example 2.4. For $\lambda = (2, 1)$ and $N = 2$:

$$\mathrm{SYT}(2,1) = \left\{ \begin{array}{|c|c|}\hline 1 & 2 \\ \hline 3 \\ \cline{1-1} \end{array}\,,\quad \begin{array}{|c|c|}\hline 1 & 3 \\ \hline 2 \\ \cline{1-1} \end{array} \right\}, \qquad \mathrm{SSYT}\big((2,1),\,2\big) = \left\{ \begin{array}{|c|c|}\hline 1 & 1 \\ \hline 2 \\ \cline{1-1} \end{array}\,,\quad \begin{array}{|c|c|}\hline 1 & 2 \\ \hline 2 \\ \cline{1-1} \end{array} \right\}.$$

The dimensions of the irreducible representations are given by:

- $\dim S^\lambda = |\operatorname{SYT}(\lambda)| = \dfrac{n!}{\prod_{u\in\lambda} h(u)}$ (hook length formula),
- $\dim W^\lambda = |\operatorname{SSYT}(\lambda, N)| = \prod_{1\leqslant i<j\leqslant N} \dfrac{\lambda_i - \lambda_j + j - i}{j-i}$ (Weyl dimension formula),

where $h(u)$ is the hook length of the box u in the Young diagram of λ.

2.3 The Schur–Weyl decomposition

Theorem 2.5 (Schur–Weyl duality)**.** *As a* (GL_N, S_n)*-bimodule,*

$$V^{\otimes n} \;\cong\; \bigoplus_{\substack{\lambda\vdash n \\ \ell(\lambda)\leqslant N}} W^\lambda \otimes S^\lambda, \tag{3}$$

where:

- W^λ *is the irreducible polynomial representation of* GL_N *of highest weight* λ *(the* Weyl module *or* Schur module*),*
- S^λ *is the irreducible representation of* S_n *indexed by* λ *(the* Specht module*).*

Moreover:

(a) The natural map $\mathbb{C}[S_n] \to \operatorname{End}_{\mathfrak{gl}_N}(V^{\otimes n})$ *is surjective (and injective when* $n \leqslant N$*).*

(b) The natural map $U(\mathfrak{gl}_N) \to \operatorname{End}_{S_n}(V^{\otimes n})$ *is surjective.*

Remark 2.6. Checking dimensions: on the left, $\dim V^{\otimes n} = N^n$. On the right,

$$\sum_{\substack{\lambda\vdash n \\ \ell(\lambda)\leqslant N}} |\operatorname{SSYT}(\lambda, N)| \cdot |\operatorname{SYT}(\lambda)| = N^n.$$

This identity can be proved bijectively via the RSK correspondence (Section 5).

3 Explicit matrix elements for S_n: Young's orthogonal form

We now describe how S_n acts on each Specht module S^λ in an explicit basis indexed by standard Young tableaux.

3.1 Content and axial distance

Definition 3.1. Let T be a standard Young tableau. The *content* of the box in row r and column c is $\operatorname{cont}(r,c) = c - r$. For each entry $k \in \{1,\dots,n\}$, we write $\operatorname{cont}_T(k)$ for the content of the box containing k in T.

The *axial distance* from k to $k{+}1$ in T is

$$\rho_T(k) = \operatorname{cont}_T(k{+}1) - \operatorname{cont}_T(k). \tag{4}$$

Example 3.2. In the standard tableau $T = \begin{array}{|c|c|}\hline 1 & 3 \\ \hline 2 \\ \cline{1-1}\end{array}$ of shape $(2,1)$: the entry 1 is in box $(1,1)$ with content 0, the entry 2 is in box $(2,1)$ with content -1, and 3 is in box $(1,2)$ with content 1. So $\rho_T(1) = \mathrm{cont}_T(2) - \mathrm{cont}_T(1) = -1 - 0 = -1$ and $\rho_T(2) = \mathrm{cont}_T(3) - \mathrm{cont}_T(2) = 1 - (-1) = 2$.

Remark 3.3. The axial distance $\rho_T(k)$ is always a nonzero integer. It is positive when $k{+}1$ appears strictly to the right and weakly below k (more precisely, when the content increases), and negative otherwise. The case $|\rho_T(k)| = 1$ occurs exactly when k and $k{+}1$ are in the same row or the same column; swapping them would violate the standard tableau condition, so $s_k \cdot T$ is undefined in that case.

3.2 The orthogonal representation

Definition 3.4. For each $\lambda \vdash n$, Young's *orthogonal representation* is the representation of S_n on the vector space $\widetilde{S}^\lambda = \bigoplus_{T \in \mathrm{SYT}(\lambda)} \mathbb{C} \cdot v_T$ defined by the action of the adjacent transpositions:

$$s_k \cdot v_T = \frac{1}{\rho_T(k)}\, v_T + \sqrt{1 - \frac{1}{\rho_T(k)^2}}\; v_{s_k \cdot T}, \tag{5}$$

where $s_k \cdot T$ is the tableau obtained from T by swapping the entries k and $k{+}1$.

If $s_k \cdot T$ is not a valid standard tableau (i.e., $|\rho_T(k)| = 1$), then the second term is absent and $s_k \cdot v_T = \pm v_T$.

Remark 3.5. There is also Young's *seminormal form* which uses the same eigenvectors but different normalization:

$$s_k \cdot v_T^{\mathrm{sn}} = \frac{1}{\rho_T(k)}\, v_T^{\mathrm{sn}} + \Big(1 + \frac{1}{\rho_T(k)}\Big) v_{s_k \cdot T}^{\mathrm{sn}}.$$

The orthogonal form has the advantage that the representing matrices are orthogonal.

Example 3.6. For $\lambda = (2,1) \vdash 3$, we have two standard tableaux:

$$T_1 = \begin{array}{|c|c|}\hline 1 & 2 \\ \hline 3 \\ \cline{1-1}\end{array}, \qquad T_2 = \begin{array}{|c|c|}\hline 1 & 3 \\ \hline 2 \\ \cline{1-1}\end{array}.$$

The axial distances are $\rho_{T_1}(1) = 1$ (so $s_1 \cdot T_1 = T_1$ is forced), $\rho_{T_1}(2) = -2$, $\rho_{T_2}(1) = -1$ (so $s_1 \cdot T_2 = -T_2$ is forced), and $\rho_{T_2}(2) = 2$. Thus in the basis (v_{T_1}, v_{T_2}):

$$s_1 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad s_2 = \begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}.$$

One can verify that $s_1^2 = s_2^2 = I$ and $(s_1 s_2)^3 = I$, confirming we have a representation of S_3.

4 Explicit matrix elements for $\mathfrak{gl}_N$: the Gel'fand–Tsetlin basis

4.1 Gel'fand–Tsetlin patterns

The branching rule for $\mathfrak{gl}_N \supset \mathfrak{gl}_{N-1}$ says that the irreducible $\mathfrak{gl}_N$-module W^λ with highest weight $\lambda = (\lambda_1, \dots, \lambda_N)$ decomposes as

$$W^\lambda\big|_{\mathfrak{gl}_{N-1}} = \bigoplus_\mu W^\mu,$$

where $\mu = (\mu_1, \dots, \mu_{N-1})$ runs over all partitions satisfying the *interlacing condition*

$$\lambda_1 \geqslant \mu_1 \geqslant \lambda_2 \geqslant \mu_2 \geqslant \cdots \geqslant \mu_{N-1} \geqslant \lambda_N.$$

Iterating this branching down the chain $\mathfrak{gl}_N \supset \mathfrak{gl}_{N-1} \supset \cdots \supset \mathfrak{gl}_1$, each basis vector is indexed by a triangular array:

Definition 4.1. A *Gel'fand–Tsetlin (GT) pattern* of shape λ (with N rows) is a triangular array

$$\Lambda = \begin{pmatrix} \lambda_{N,1} & & \lambda_{N,2} & & \cdots & & \lambda_{N,N} \\ & \lambda_{N-1,1} & & \lambda_{N-1,2} & & \cdots & \lambda_{N-1,N-1} \\ & & \ddots & & & & \\ & & & \lambda_{1,1} & & & \end{pmatrix}$$

with $(\lambda_{N,1}, \dots, \lambda_{N,N}) = \lambda$ and satisfying the interlacing conditions $\lambda_{k,i} \geqslant \lambda_{k-1,i} \geqslant \lambda_{k,i+1}$ for all valid indices.

Remark 4.2. The set of GT patterns of shape λ is in natural bijection with $\mathrm{SSYT}(\lambda, N)$: given a semistandard tableau T, the k-th row of the GT pattern records the shape of the subtableau of T consisting of entries $\leqslant k$.

4.2 The $\mathfrak{gl}_N$ action in the GT basis

Write v_Λ for the basis vector of W^λ indexed by the GT pattern Λ. The Lie algebra $\mathfrak{gl}_N$ is generated by the matrix units $E_{k,k}$ (Cartan subalgebra) and $E_{k,k+1}$, $E_{k+1,k}$ (raising and lowering operators).

Proposition 4.3 (Gel'fand–Tsetlin formulae)**.** *The action of $\mathfrak{gl}_N$ on the GT basis is:*

$$E_{k,k}\, v_\Lambda = \left(\sum_{i=1}^{k} \lambda_{k,i} - \sum_{i=1}^{k-1} \lambda_{k-1,i} \right) v_\Lambda, \tag{6}$$

$$E_{k,k+1}\, v_\Lambda = -\sum_{i=1}^{k} \frac{\prod_{j=1}^{k+1} (\lambda_{k,i} - \lambda_{k+1,j} + j - i) \prod_{j=1}^{k-1} (\lambda_{k,i} - \lambda_{k-1,j} + j - i)}{\prod_{j \neq i}^{k} (\lambda_{k,i} - \lambda_{k,j} + j - i)}\, v_{\Lambda + \delta_{k,i}}, \tag{7}$$

$$E_{k+1,k}\, v_\Lambda = \sum_{i=1}^{k} \frac{\prod_{j=1}^{k+1} (\lambda_{k,i} - \lambda_{k+1,j} + j - i) \prod_{j=1}^{k-1} (\lambda_{k,i} - \lambda_{k-1,j} + j - i)}{\prod_{j \neq i}^{k} (\lambda_{k,i} - \lambda_{k,j} + j - i)}\, v_{\Lambda - \delta_{k,i}}, \tag{8}$$

where $\Lambda \pm \delta_{k,i}$ denotes the GT pattern obtained by replacing $\lambda_{k,i}$ with $\lambda_{k,i} \pm 1$ (and the term is zero if the result is not a valid GT pattern).

The eigenvalue in (6) is the weight: $E_{k,k}$ counts how many entries equal to k appear in the corresponding semistandard tableau.

Remark 4.4 (Shifted coordinates). The GT formulae simplify in *shifted coordinates* $\ell_{k,i} = \lambda_{k,i} - i$, which is the content of the rightmost box in row i of the partition $\lambda^{(k)} = (\lambda_{k,1}, \dots, \lambda_{k,k})$. The interlacing condition becomes $\ell_{k+1,j} \geqslant \ell_{k,j} \geqslant \ell_{k+1,j+1}$, and every factor in the raising and lowering formulae becomes a difference of shifted entries:

$$E_{k,k+1}\, v_\Lambda = -\sum_{i=1}^{k} \frac{\displaystyle\prod_{j=1}^{k+1}(\ell_{k,i} - \ell_{k+1,j}) \prod_{j=1}^{k-1}(\ell_{k,i} - \ell_{k-1,j})}{\displaystyle\prod_{\substack{j=1 \\ j\neq i}}^{k}(\ell_{k,i} - \ell_{k,j})}\, v_{\Lambda+\delta_{k,i}}, \tag{9}$$

$$E_{k+1,k}\, v_\Lambda = \sum_{i=1}^{k} \frac{\displaystyle\prod_{j=1}^{k+1}(\ell_{k,i} - \ell_{k+1,j}) \prod_{j=1}^{k-1}(\ell_{k,i} - \ell_{k-1,j})}{\displaystyle\prod_{\substack{j=1 \\ j\neq i}}^{k}(\ell_{k,i} - \ell_{k,j})}\, v_{\Lambda-\delta_{k,i}}. \tag{10}$$

The structure of each coefficient is now transparent:

- **Numerator**: product of distances from $\ell_{k,i}$ to the shifted entries on the rows immediately above $(\ell_{k+1,1}, \dots, \ell_{k+1,k+1})$ and below $(\ell_{k-1,1}, \dots, \ell_{k-1,k-1})$ in the GT pattern.
- **Denominator**: Vandermonde-type product of distances between $\ell_{k,i}$ and all other shifted entries on the same row k.

The Weyl dimension formula takes the same shape: $\dim W^\lambda = \prod_{1\leqslant i<j\leqslant N} \frac{\ell_i - \ell_j}{j-i}$, where $\ell_i = \lambda_i - i$. The GT coefficients are *local* versions of this formula: where the dimension involves all pairs of shifted entries across the full partition, each GT coefficient involves only the entry being moved and its neighbours in adjacent rows.

5 The Robinson–Schensted–Knuth correspondence

We now describe the bijection that ties everything together.

5.1 Row insertion

Given a semistandard tableau P and a positive integer x, the *row insertion* $P \leftarrow x$ produces a new semistandard tableau by the following algorithm:

1. Start with the first row of P.
2. If x is weakly larger than all entries in the current row, place x at the end of the row. Terminate.
3. Otherwise, find the leftmost entry y in the current row with $y > x$. Replace y by x ("x bumps y").
4. Repeat from step 2 with y and the next row down.

The key property is that insertion preserves the semistandard condition.

5.2 The RSK correspondence

Definition 5.1. The *Robinson–Schensted–Knuth (RSK) correspondence* is a bijection

$$\mathrm{RSK} \;:\; \{1,\dots,N\}^n \;\xrightarrow{\sim}\; \bigsqcup_{\substack{\lambda \vdash n \\ \ell(\lambda) \leqslant N}} \mathrm{SSYT}(\lambda, N) \times \mathrm{SYT}(\lambda)$$

defined as follows. Given a word $\mathbf{i} = (i_1, \dots, i_n)$, construct a pair of tableaux (P, Q) by iterating:

- Start with $P_0 = Q_0 = \varnothing$ (empty tableaux).
- At step $k = 1, \dots, n$: insert i_k into P_{k-1} to get $P_k = P_{k-1} \leftarrow i_k$. This insertion adds a new box to the shape; place k in that box to form Q_k.

Set $P = P_n$ (the *insertion tableau*) and $Q = Q_n$ (the *recording tableau*).

Example 5.2. Let $N = 2$, $n = 3$, and $\mathbf{i} = (2, 1, 1)$.

- Insert 2: $P_1 = \begin{array}{|c|}\hline 2 \\ \hline \end{array}$, $\quad Q_1 = \begin{array}{|c|}\hline 1 \\ \hline \end{array}$.
- Insert 1: 1 bumps 2 from row 1, 2 goes to row 2. $P_2 = \begin{array}{|c|}\hline 1 \\ \hline 2 \\ \hline \end{array}$, $\quad Q_2 = \begin{array}{|c|}\hline 1 \\ \hline 2 \\ \hline \end{array}$.
- Insert 1: we bump the leftmost entry *strictly* greater than 1 in row 1. There is no such entry (the only entry is 1 itself), so 1 appends to the end of row 1. $P_3 = \begin{array}{|c|c|}\hline 1 & 1 \\ \hline 2 \\ \cline{1-1} \end{array}$, $\quad Q_3 = \begin{array}{|c|c|}\hline 1 & 3 \\ \hline 2 \\ \cline{1-1} \end{array}$.

So $\mathrm{RSK}(2,1,1) = (P, Q)$ with $P = \begin{array}{|c|c|}\hline 1 & 1 \\ \hline 2 \\ \cline{1-1} \end{array}$ and $Q = \begin{array}{|c|c|}\hline 1 & 3 \\ \hline 2 \\ \cline{1-1} \end{array}$, both of shape $\lambda = (2, 1)$.

5.3 Properties of RSK

Proposition 5.3. *The RSK correspondence has the following properties:*

(a) P is a semistandard Young tableau with entries in $\{1, \dots, N\}$.

(b) Q is a standard Young tableau.

(c) $\mathrm{shape}(P) = \mathrm{shape}(Q) = \lambda$ *for some partition $\lambda \vdash n$ with $\ell(\lambda) \leqslant N$.*

(d) The correspondence is a bijection.

(e) The weight *of P (the tuple recording how many times each letter appears) equals the weight of the word $\mathbf{i}$; specifically, the number of entries equal to a in P equals the number of occurrences of a in $\mathbf{i}$.*

6 RSK is the Schur–Weyl decomposition

We now state the main interpretive result: the RSK correspondence is not merely a combinatorial bijection, but the *explicit realisation* of the Schur–Weyl decomposition (3).

6.1 The decomposition, concretely

The tensor product basis element $e_{\mathbf{i}} \in V^{\otimes n}$ belongs to the summand $W^\lambda \otimes S^\lambda$ in (3), where $\lambda = \mathrm{shape}(P) = \mathrm{shape}(Q)$ and $(P, Q) = \mathrm{RSK}(\mathbf{i})$. More precisely:

Theorem 6.1 (RSK as Schur–Weyl)**.** *Under the isomorphism* (3)*, the basis element* $e_{\mathbf{i}}$ *corresponds to*

$$e_{\mathbf{i}} \;\longleftrightarrow\; w_P \otimes v_Q \;\in\; W^\lambda \otimes S^\lambda,$$

where:

- $\lambda = \mathrm{shape}(\mathrm{RSK}(\mathbf{i}))$ *determines which summand,*
- $P = P(\mathbf{i})$*, the insertion tableau, gives coordinates in the* GL_N*-irrep* W^λ *(via the GT-basis/SSYT-basis identification),*
- $Q = Q(\mathbf{i})$*, the recording tableau, gives coordinates in the* S_n*-irrep* S^λ *(via the SYT basis).*

Remark 6.2. The correspondence is not an isometry with respect to the naive inner products—the tensor product basis is orthonormal while the RSK correspondence identifies it with a *different* basis of $\bigoplus_\lambda W^\lambda \otimes S^\lambda$. The precise relationship involves the transition matrix between the tensor product basis and the Schur–Weyl adapted basis, which can be computed recursively from the RSK insertion algorithm.

6.2 What each part of RSK "sees"

The beauty of this interpretation is that the two tableaux capture complementary information:

	Insertion tableau P	**Recording tableau** Q
Type	Semistandard (SSYT)	Standard (SYT)
Entries	Letters from $\{1, \ldots, N\}$	Positions $\{1, \ldots, n\}$
Indexes	GL_N-irrep basis	S_n-irrep basis
Sensitive to	What letters appear	In what order they arrive
Dimension count	$\lvert\,\mathrm{SSYT}(\lambda, N)\rvert = \dim W^\lambda$	$\lvert\,\mathrm{SYT}(\lambda)\rvert = \dim S^\lambda$
Symmetry	Changed by GL_N action	Changed by S_n action

6.3 How the two actions transform (P, Q)

Proposition 6.3. *Under the RSK identification:*

(a) The S_n*-action permutes the recording tableau* Q *(changing the* S_n *coordinates) while preserving the insertion tableau* P *(the* GL_N *coordinates are invisible to* S_n*).*

(b) The GL_N*-action transforms the insertion tableau* P *(changing the* GL_N *coordinates) while preserving the recording tableau* Q *(the* S_n *coordinates are invisible to* GL_N*).*

This is exactly the content of Schur–Weyl duality: the two actions commute because they act on complementary factors of $W^\lambda \otimes S^\lambda$*.*

Remark 6.4. Statement (a) requires care: S_n does not simply permute entries of Q in a naive sense. Rather, the RSK correspondence intertwines the S_n-action on words with a well-defined action on standard tableaux that corresponds to the representation of S_n on S^λ. The precise statement involves the *Knuth equivalence*: two words $\mathbf{i}$ and $\mathbf{j}$ have the same insertion tableau P if and only if they are Knuth-equivalent.

6.4 The dimension identity

The bijectivity of RSK immediately gives the counting identity:

$$N^n = |\{1,\dots,N\}^n| = \sum_{\substack{\lambda\vdash n\\ \ell(\lambda)\leqslant N}} |\operatorname{SSYT}(\lambda,N)|\cdot|\operatorname{SYT}(\lambda)| = \sum_{\substack{\lambda\vdash n\\ \ell(\lambda)\leqslant N}} \dim W^\lambda\cdot\dim S^\lambda, \quad (11)$$

which is the dimension check for the Schur–Weyl decomposition (3).

Example 6.5. For $N=2$, $n=3$: we need partitions of 3 with at most 2 parts, namely $\lambda=(3)$ and $\lambda=(2,1)$.

- $\lambda=(3)$: $|\operatorname{SSYT}((3),2)| = |\{111,112,122,222\}| = 4$, $|\operatorname{SYT}((3))| = 1$. Contribution: $4\cdot 1 = 4$.
- $\lambda=(2,1)$: $|\operatorname{SSYT}((2,1),2)| = 2$ (see Example 2.4), $|\operatorname{SYT}((2,1))| = 2$. Contribution: $2\cdot 2 = 4$.

Total: $4+4=8=2^3$. ✓

7 The bridge: Jucys–Murphy elements and spectral projectors

The preceding sections construct the S_n and $\mathfrak{gl}_N$ matrix elements *intrinsically*: Young's orthogonal form (Section 3) uses the combinatorics of standard tableaux, and the GT formulae (Section 4) use the branching chain $\mathfrak{gl}_N \supset \mathfrak{gl}_{N-1} \supset \cdots \supset \mathfrak{gl}_1$. Section 6 claims RSK provides the change of basis, but never makes the embedding explicit. How do these abstract representations actually sit inside $V^{\otimes n}$?

The missing ingredient is the *Jucys–Murphy elements*, which provide concrete projection operators from $V^{\otimes n}$ onto each irreducible summand.

7.1 Motivation: building a commutative subalgebra

How would one construct a large commutative subalgebra of $\mathbb{C}[S_n]$?

Strategy: use the chain $S_1 \subset S_2 \subset \cdots \subset S_n$. The centre $Z(\mathbb{C}[S_k])$ commutes with *everything* in $\mathbb{C}[S_k]$, and in particular with $Z(\mathbb{C}[S_j])$ for $j \leqslant k$ (since $\mathbb{C}[S_j] \subset \mathbb{C}[S_k]$). So the algebra generated by all the centres $Z(\mathbb{C}[S_1]), Z(\mathbb{C}[S_2]), \dots, Z(\mathbb{C}[S_n])$ is commutative. This is the *Gelfand–Tsetlin (GZ) subalgebra*.

Each centre $Z(\mathbb{C}[S_k])$ is spanned by class sums. At step k, the "new" class sum not already in $\mathbb{C}[S_{k-1}]$ is $X_k = \sum_{1\leqslant i<j\leqslant k}(i\ j)$ (the sum of all transpositions in S_k), which is central in $\mathbb{C}[S_k]$. Set $J_k = X_k - X_{k-1}$: this peels off the contribution of the new element k, and is the sum of all transpositions that *involve* k.

The element J_k measures how k interacts with the earlier elements $\{1, \dots, k{-}1\}$. Since these interactions at different steps k and l are mediated by disjoint "new arrivals," one might hope they commute—and indeed they do. More is true: the symmetric power sums $J_1^r + \cdots + J_k^r$ are central in $\mathbb{C}[S_k]$ for every r, so $J_1, \dots, J_n$ actually generate the *entire* GZ subalgebra, not just a piece of it [8].

7.2 The Jucys–Murphy elements

Definition 7.1. The *Jucys–Murphy elements* of $\mathbb{C}[S_n]$ are

$$J_1 = 0, \qquad J_k = (1\ k) + (2\ k) + \cdots + (k{-}1\ \ k), \quad k = 2, \dots, n.$$

Equivalently, $J_k = X_k - X_{k-1}$ where $X_k = \sum_{1 \leqslant i < j \leqslant k} (i\ j) \in Z(\mathbb{C}[S_k])$.

Lemma 7.2. *The elements $J_1, \dots, J_n$ commute pairwise.*

Proof. Let $k < l$. We compute $[J_k, J_l] = \sum_{i<k} \sum_{j<l} [(i\ k),\ (j\ l)]$ by sorting the terms of the double sum according to whether the two transpositions share an index.

Case 1: $\{i, k\} \cap \{j, l\} = \varnothing$. The transpositions $(i\ k)$ and $(j\ l)$ are disjoint, hence commute. Contribution: 0.

Case 2: $i = j$ *(so* $i < k < l$*).* We get $[(i\ k),\ (i\ l)] = (i\ k\ l) - (i\ l\ k)$.

Case 3: $j = k$ *(so* $i < k < l$*).* We get $[(i\ k),\ (k\ l)] = (i\ l\ k) - (i\ k\ l)$.

No other overlaps occur: $i = l$ is impossible since $i < k < l$, and $j = k$ with $j = i$ would force $i = k$.

For each value of $i < k$, the Case 2 and Case 3 contributions sum to

$$\big((i\ k\ l) - (i\ l\ k)\big) + \big((i\ l\ k) - (i\ k\ l)\big) = 0.$$

So $[J_k, J_l] = 0$. □

The subalgebra of $\mathbb{C}[S_n]$ generated by $J_1, \dots, J_n$ is called the *Gelfand–Tsetlin (GZ) subalgebra* for the chain $S_1 \subset S_2 \subset \cdots \subset S_n$ [8]. By "commutative subalgebra" we mean that the J_k commute *with each other*—they certainly do not commute with arbitrary elements of $\mathbb{C}[S_n]$.

Lemma 7.3. *For $k \geqslant 2$,*

$$J_k = s_{k-1}\, J_{k-1}\, s_{k-1} + s_{k-1}. \tag{12}$$

Proof. We show $s_{k-1}\, J_{k-1}\, s_{k-1} = J_k - s_{k-1}$. Recall $J_{k-1} = \sum_{i=1}^{k-2} (i,\ k{-}1)$ and $s_{k-1} = (k{-}1,\ k)$. Conjugating each summand:

$$s_{k-1}\,(i,\ k{-}1)\,s_{k-1} = (k{-}1,\ k)\,(i,\ k{-}1)\,(k{-}1,\ k) = (i,\ k), \qquad i < k{-}1,$$

since conjugation replaces every occurrence of $k{-}1$ by k. Summing over $i = 1, \dots, k{-}2$:

$$s_{k-1}\, J_{k-1}\, s_{k-1} = \sum_{i=1}^{k-2} (i,\ k) = J_k - (k{-}1,\ k) = J_k - s_{k-1}.$$ □

Proposition 7.4. *In the orthogonal representation $\widetilde{S}^\lambda$ (Definition 3.4), the Jucys–Murphy element J_k acts diagonally in the SYT basis:*

$$J_k \cdot v_T = \mathrm{cont}_T(k)\, v_T,$$

where $\mathrm{cont}_T(k) = c - r$ is the content of the box in row r, column c that contains entry k in T (Definition 3.1).

Proof. We use the recursion of Lemma 7.3 and induction on k.

Base case ($k = 1$): $J_1 = 0$, and entry 1 is always in box $(1, 1)$ with content 0.

Inductive step: Assume $J_{k-1}\, v_T = \mathrm{cont}_T(k{-}1)\, v_T$ for all T. Set $\rho = \rho_T(k{-}1) = \mathrm{cont}_T(k) - \mathrm{cont}_T(k{-}1)$, and write $\alpha = 1/\rho$ and $\beta = \sqrt{1 - 1/\rho^2}$. If $|\rho| = 1$ then $\beta = 0$ and $s_{k-1}\, v_T = \alpha\, v_T$; the computation below applies with $\beta = 0$. Otherwise let $T' = s_{k-1} \cdot T$ (the tableau with entries $k{-}1$ and k swapped). Note that $\mathrm{cont}_{T'}(k{-}1) = \mathrm{cont}_T(k)$ and $\rho_{T'}(k{-}1) = -\rho$, so from Young's orthogonal form:

$$s_{k-1}\, v_T = \alpha\, v_T + \beta\, v_{T'}, \qquad s_{k-1}\, v_{T'} = -\alpha\, v_{T'} + \beta\, v_T.$$

Now compute $s_{k-1}\, J_{k-1}\, s_{k-1}\, v_T$ step by step. Writing $c_1 = \mathrm{cont}_T(k{-}1)$ and $c_2 = \mathrm{cont}_T(k)$:

$$\begin{aligned} s_{k-1}\, v_T &= \alpha\, v_T + \beta\, v_{T'}, \\ J_{k-1}(\alpha\, v_T + \beta\, v_{T'}) &= \alpha\, c_1\, v_T + \beta\, c_2\, v_{T'} \qquad \text{(by induction)}, \\ s_{k-1}(\alpha\, c_1\, v_T + \beta\, c_2\, v_{T'}) &= \alpha\, c_1\, (\alpha\, v_T + \beta\, v_{T'}) + \beta\, c_2\, (\beta\, v_T - \alpha\, v_{T'}) \\ &= \big(\alpha^2 c_1 + \beta^2 c_2\big)\, v_T + \alpha\beta\big(c_1 - c_2\big)\, v_{T'}. \end{aligned}$$

Since $c_1 - c_2 = -\rho$ and $\alpha = 1/\rho$:

- the $v_{T'}$ coefficient is $\alpha\beta(-\rho) = -\beta$,
- the v_T coefficient is $\alpha^2 c_1 + (1 - \alpha^2)\, c_2 = c_2 + \alpha^2(c_1 - c_2) = c_2 - 1/\rho$.

Adding $s_{k-1}\, v_T = (1/\rho)\, v_T + \beta\, v_{T'}$:

$$J_k\, v_T = \big(c_2 - \tfrac{1}{\rho} + \tfrac{1}{\rho}\big)\, v_T + (-\beta + \beta)\, v_{T'} = c_2\, v_T = \mathrm{cont}_T(k)\, v_T.$$

The off-diagonal terms cancel exactly. □

That is, the SYT basis of Young's orthogonal form is precisely the *simultaneous eigenbasis* of all Jucys–Murphy elements. The *content vector* of a standard tableau T is $\mathbf{c}(T) = (\mathrm{cont}_T(1), \ldots, \mathrm{cont}_T(n))$.

Remark 7.5 (What simultaneous diagonalisation achieves). We first recall a general fact from linear algebra.

The single-operator case. Let A be a diagonalisable operator on a finite-dimensional vector space V with distinct eigenvalues $c_1, \ldots, c_m$. For each i, define

$$P_i = \prod_{j \neq i} \frac{A - c_j}{c_i - c_j}. \tag{13}$$

This is a polynomial in A of degree $m - 1$. We claim P_i is the projector onto the c_i-eigenspace.

Proof. Let v be an eigenvector with $Av = c_k v$. Substituting $A = c_k$ into (13):

$$P_i\, v = \prod_{j \neq i} \frac{c_k - c_j}{c_i - c_j}\, v.$$

If $k = i$, every factor is $(c_i - c_j)/(c_i - c_j) = 1$, so $P_i\, v = v$. If $k \neq i$, the factor $j = k$ gives $(c_k - c_k)/(c_i - c_k) = 0$, so $P_i\, v = 0$. Since eigenvectors span V, this determines P_i completely: it acts as the identity on the c_i-eigenspace and as zero on all other eigenspaces.

In particular: $P_i^2 = P_i$ (applying the projector twice is the same as once), $P_i P_j = 0$ for $i \neq j$ (projecting onto c_i then c_j kills everything), and $\sum_{i=1}^{m} P_i = I$ (every vector belongs to some eigenspace). Furthermore, A itself is recovered as $A = \sum_{i=1}^{m} c_i P_i$ (the *spectral decomposition*).

Multiple commuting operators. If $A_1, \ldots, A_n$ are simultaneously diagonalisable, the projector onto the joint eigenspace with eigenvalue tuple $(c_1, \ldots, c_n)$ is the product of the individual Lagrange projectors:

$$P_{(c_1,\ldots,c_n)} = \prod_{k=1}^{n} \prod_{\substack{c \neq c_k \\ c \text{ an eigenvalue of } A_k}} \frac{A_k - c}{c_k - c}.$$

This is a polynomial in $A_1, \ldots, A_n$; it acts as the identity on the target eigenspace and as zero on all others, so $P^2 = P$.

For the Jucys–Murphy elements, this general procedure has a remarkable consequence. The operators $J_1, \ldots, J_n$ live in $\mathbb{C}[S_n]$, so their Lagrange projectors are elements of $\mathbb{C}[S_n]$ too—they are the primitive idempotents e_Q of Definition 7.8. Each eigenvalue tuple $\mathbf{c}(T)$ is the content vector of a standard tableau T, and the content vector carries two levels of information:

- The *multiset* of contents $\{\mathrm{cont}_T(1), \ldots, \mathrm{cont}_T(n)\}$ determines the shape λ (which irreducible representation).
- The *sequence* of contents $(\mathrm{cont}_T(1), \ldots, \mathrm{cont}_T(n))$ determines the specific tableau T (which basis vector within that irreducible).

So the joint eigenspace decomposition is *finer* than the irreducible decomposition: each irreducible S^λ of dimension $d_\lambda = |\mathrm{SYT}(\lambda)|$ splits into d_λ one-dimensional eigenspaces, one per standard tableau of shape λ. Grouping eigenspaces by shape recovers the irreducible decomposition.

Applied to *any* representation V of S_n: the idempotents $e_Q \in \mathbb{C}[S_n]$ act on V, and the subspaces $e_Q \cdot V$ decompose V into pieces labelled by standard tableaux. This is one diagonalisation, performed once inside the algebra, valid for all representations simultaneously.

Lemma 7.6. *Different standard tableaux have different content vectors: if T and T' are standard tableaux (of any shapes) with $\mathbf{c}(T) = \mathbf{c}(T')$, then $T = T'$.*

Proof. A standard tableau T is built step by step: at step k, the entry k is placed in an addable corner of the shape $\lambda^{(k-1)}$ (the shape of entries $1, \ldots, k-1$) to produce $\lambda^{(k)}$. The content $\mathrm{cont}_T(k)$ records *which* corner was chosen.

The key observation: the addable corners of any Young diagram have distinct contents (each lies on a different diagonal $c - r$). So knowing $\mathrm{cont}_T(k)$ and the shape $\lambda^{(k-1)}$ determines the shape $\lambda^{(k)}$ uniquely.

Since $\lambda^{(0)} = \varnothing$ for every SYT, induction on k shows that the content vector $(\mathrm{cont}_T(1), \ldots, \mathrm{cont}_T(n))$ determines the entire sequence of shapes $\varnothing = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(n)}$, which is the tableau T. □

Theorem 7.7. *The GZ subalgebra is a maximal commutative subalgebra of $\mathbb{C}[S_n]$.*

Proof. By Artin–Wedderburn, $\mathbb{C}[S_n] \cong \bigoplus_{\lambda \vdash n} \mathrm{Mat}_{d_\lambda}(\mathbb{C})$ where $d_\lambda = |\operatorname{SYT}(\lambda)|$. A commutative subalgebra of $\mathrm{Mat}_d(\mathbb{C})$ has dimension at most d (since commuting matrices can be simultaneously upper-triangularised, and diagonal matrices in that basis span a d-dimensional commutative subalgebra). So any commutative subalgebra of $\mathbb{C}[S_n]$ has dimension at most $\sum_{\lambda \vdash n} d_\lambda$.

By Proposition 7.4, the GZ subalgebra acts on each $\widetilde{S}^\lambda$ by diagonal matrices in the SYT basis $\{v_T : T \in \operatorname{SYT}(\lambda)\}$. By Lemma 7.6, different basis vectors have different joint eigenvalues, so the GZ subalgebra contains *all* diagonal matrices in this basis (any diagonal matrix is a polynomial in commuting operators with distinct joint eigenvalues—use Lagrange interpolation). Thus the GZ subalgebra contains a d_λ-dimensional subalgebra in each block, giving total dimension at least $\sum_\lambda d_\lambda$.

Since the GZ subalgebra is commutative (Lemma 7.2) and achieves the maximum possible dimension for a commutative subalgebra, it is maximal. □

7.3 Spectral projectors

Since the J_k are simultaneously diagonalisable, we can build projection operators from their spectral decomposition.

Definition 7.8. For each $Q \in \operatorname{SYT}(\lambda)$, the *primitive idempotent* is

$$e_Q \;=\; \prod_{k=2}^{n} \; \prod_{\substack{c \neq \mathrm{cont}_Q(k) \\ c \text{ an eigenvalue of } J_k}} \frac{J_k - c}{\mathrm{cont}_Q(k) - c} \;\in\; \mathbb{C}[S_n].$$

These idempotents satisfy:

(i) $e_Q^2 = e_Q$ and $e_Q\, e_{Q'} = 0$ for $Q \neq Q'$;

(ii) $\displaystyle\sum_Q e_Q = 1$, the sum over all SYT of all shapes;

(iii) $c_\lambda := \displaystyle\sum_{Q \in \operatorname{SYT}(\lambda)} e_Q$ is the central idempotent projecting onto the λ-isotypic component.

Remark 7.9. Alternatively, c_λ can be expressed via characters:

$$c_\lambda = \frac{\dim S^\lambda}{n!} \sum_{\sigma \in S_n} \chi^\lambda(\sigma)\, \sigma,$$

but the Jucys–Murphy construction has the advantage of producing the *primitive* idempotents e_Q directly, without requiring knowledge of the full character table.

7.4 Carving out the Weyl modules

Since $e_Q \in \mathbb{C}[S_n]$ and the $\mathfrak{gl}_N$-action commutes with S_n (Proposition 1.1), the subspace $e_Q \cdot V^{\otimes n} \subset V^{\otimes n}$ is a $\mathfrak{gl}_N$-submodule. This is the bridge:

Proposition 7.10. *For each $Q \in \operatorname{SYT}(\lambda)$ with $\ell(\lambda) \leqslant N$:*

(a) $e_Q \cdot V^{\otimes n}$ *is an irreducible* $\mathfrak{gl}_N$*-module isomorphic to* W^λ*.*

(b) Under this isomorphism, the GT basis vector v_Λ *corresponds to* $e_Q \cdot e_{\mathbf{i}}$*, where* $\mathbf{i}$ *is any word with* $\mathrm{RSK}(\mathbf{i}) = (P, Q)$ *and* P *is the SSYT corresponding to* Λ*.*

(c) Varying Q *over* $\mathrm{SYT}(\lambda)$ *gives the* $\dim S^\lambda$ *distinct copies of* W^λ *constituting the* λ*-isotypic component of* $V^{\otimes n}$*.*

This resolves the gap identified in Remark 6.2: the isomorphism (3) is realised concretely by $e_{\mathbf{i}} \mapsto \sum_Q e_Q \cdot e_{\mathbf{i}}$, decomposing each tensor into its projections onto the copies of W^λ labelled by standard tableaux.

7.5 How the two sets of matrix elements act on $V^{\otimes n}$

The explicit formulae of Sections 3 and 4 now have a concrete home:

- **The GT formulae** (Proposition 4.3) describe the $\mathfrak{gl}_N$-action *within* each projected subspace $e_Q \cdot V^{\otimes n}$, for a fixed Q.
- **Young's orthogonal form** (equation (5)) describes how the S_n-action *rotates between* different copies: a transposition s_k sends $e_Q \cdot V^{\otimes n}$ to a linear combination of $e_Q \cdot V^{\otimes n}$ and $e_{s_k \cdot Q} \cdot V^{\otimes n}$, with coefficients given by (5).

7.6 Duality of branching chains

The two sets of matrix elements arise from two dual branching chains acting on orthogonal aspects of the tensor space:

	S_n **chain**	$\mathfrak{gl}_N$ **chain**
Chain	$S_1 \subset S_2 \subset \cdots \subset S_n$	$\mathfrak{gl}_1 \subset \mathfrak{gl}_2 \subset \cdots \subset \mathfrak{gl}_N$
Acts on	Positions $\{1, \ldots, n\}$	Values $\{1, \ldots, N\}$
On $V^{\otimes n}$	Permutes which slot each value occupies	Mixes which value each slot contains
Branching gives	SYT basis (growth of shape by position)	GT basis (growth of shape by value)
GZ generators	Jucys–Murphy elements J_k	Cartan elements of $\mathfrak{gl}_k$
Eigenvalues	Contents $\mathrm{cont}_T(k)$	Weights $\sum_i \lambda_{k,i} - \sum_i \lambda_{k-1,i}$

These two branching structures commute because positions and values are independent: permuting tensor positions does not interfere with mixing basis vectors within each position. The Schur–Weyl decomposition is the *simultaneous spectral decomposition* with respect to both commuting families of operators.

8 Connection to the categorical framework

Following Davydov–Molev [1], the classical Schur–Weyl duality can be reformulated categorically as follows.

The sequence of group algebras $k[S_*] = \{k[S_n] \mid n \geqslant 0\}$ with the natural maps $\mu_{m,n} : k[S_m] \otimes k[S_n] \to k[S_{m+n}]$ induced by $S_m \times S_n \hookrightarrow S_{m+n}$ forms a *multiplicative sequence of*

algebras. The associated *Schur–Weyl category* $\mathcal{S} = \mathcal{C}(k[S_*])$ is a free symmetric monoidal category generated by one object, with endomorphism algebras $\mathrm{End}_{\mathcal{S}}(X^{\otimes n}) = k[S_n]$.

The *symmetric fiber functor* $F_N : \mathcal{S} \to \mathcal{V}ect$ labelled by N sends the generator X to $V = k^N$, and on morphisms gives the map $k[S_n] \to \mathrm{End}(V^{\otimes n})$. This functor factors through the category of representations of $\mathfrak{gl}_N$ via the *Schur–Weyl functor* $SW_N : \mathcal{S} \to \mathcal{R}\mathrm{ep}(\mathfrak{gl}_N)$.

The classical Schur–Weyl duality (Theorem 2.5) is precisely the statement that SW_N is *full*: every $\mathfrak{gl}_N$-equivariant map between tensor powers of V comes from the symmetric group. The RSK correspondence provides the explicit combinatorial witness for this fullness, by decomposing $V^{\otimes n}$ into irreducible bimodule summands.

The power of the Davydov–Molev framework is that the same pattern applies to quantum and affine deformations: replacing $k[S_*]$ by the Hecke algebras $\mathrm{H}_*(q)$ gives the Hecke category whose fiber functors factor through $U_q(\mathfrak{gl}_N)$-modules; replacing by degenerate affine Hecke algebras gives functors to Yangian modules; and replacing by affine Hecke algebras gives functors to quantum affine algebra modules. In each case, the universal property of the Schur–Weyl category reduces the duality to a statement about generators and relations.

Part II
q-Deformed Schur–Weyl Duality

Throughout Part II, we fix a parameter $q \in \mathbb{C}^*$ with q not a root of unity. All constructions reduce to the classical case of Part I when $q \to 1$.

We use the notation

$$[x]_q = \frac{q^x - q^{-x}}{q - q^{-1}}$$

for the *q-number*. When x is a positive integer n, this is the *q-integer* $[n]_q = q^{n-1} + q^{n-3} + \cdots + q^{-(n-1)}$. Note $[x]_q \to x$ as $q \to 1$.

9 The two commuting actions (q-deformed)

Fix $V = \mathbb{C}^N$ with standard basis $e_1, \ldots, e_N$, as in Part I. The tensor power $V^{\otimes n}$ has the same basis $\{e_{\mathbf{i}}\}$, but the two group actions of Part I are replaced by algebraic actions that deform the GL_N and S_n symmetries.

9.1 The quantum group $U_q(\mathfrak{gl}_N)$

Definition 9.1. The *quantum group* $U_q(\mathfrak{gl}_N)$ is the $\mathbb{C}$-algebra generated by E_i, F_i $(1 \leqslant i \leqslant N-1)$ and $K_j^{\pm 1}$ $(1 \leqslant j \leqslant N)$ subject to:

(i) $K_i K_j = K_j K_i, \quad K_i K_i^{-1} = 1;$

(ii) $K_i E_j K_i^{-1} = q^{\delta_{ij} - \delta_{i,j+1}} E_j;$

(iii) $K_i F_j K_i^{-1} = q^{-\delta_{ij} + \delta_{i,j+1}} F_j;$

(iv) $E_iF_j - F_jE_i = \delta_{ij}\dfrac{K_iK_{i+1}^{-1} - K_i^{-1}K_{i+1}}{q-q^{-1}}$;

(v) the q-Serre relations: $E_i^2E_j - (q+q^{-1})E_iE_jE_i + E_jE_i^2 = 0$ for $|i-j| = 1$ (and similarly for F_i).

The generators K_i form a *quantum Cartan subalgebra*; E_i and F_i are *quantum raising and lowering operators*.

At $q = 1$, setting $K_i = 1$ for all i and $E_i \to E_{i,i+1}$, $F_i \to E_{i+1,i}$, we recover the universal enveloping algebra $U(\mathfrak{gl}_N)$.

Definition 9.2. $U_q(\mathfrak{gl}_N)$ acts on $V = \mathbb{C}^N$ by:

$$K_j \cdot e_a = q^{\delta_{ja}}\, e_a, \qquad E_i \cdot e_a = \delta_{a,i+1}\, e_i, \qquad F_i \cdot e_a = \delta_{a,i}\, e_{i+1}.$$

This is the *vector representation*.

The action on $V^{\otimes n}$ uses the *coproduct* $\Delta : U_q(\mathfrak{gl}_N) \to U_q(\mathfrak{gl}_N) \otimes U_q(\mathfrak{gl}_N)$:

$$\Delta(K_i) = K_i \otimes K_i, \qquad \Delta(E_i) = E_i \otimes K_iK_{i+1}^{-1} + 1 \otimes E_i, \qquad \Delta(F_i) = F_i \otimes 1 + K_i^{-1}K_{i+1} \otimes F_i.$$

Iterating Δ gives an action of $U_q(\mathfrak{gl}_N)$ on $V^{\otimes n}$.

Remark 9.3. Unlike the classical case, this action is *not* diagonal: E_i acting on the k-th tensor factor inserts a factor of $K_iK_{i+1}^{-1}$ on all factors to its left. This asymmetry is the price of deformation, and it is precisely what the R-matrix corrects.

9.2 The Hecke algebra $\mathrm{H}_n(q)$

Definition 9.4. The *Iwahori–Hecke algebra* $\mathrm{H}_n(q)$ is the $\mathbb{C}$-algebra generated by $T_1, \dots, T_{n-1}$ subject to:

(i) *Quadratic relation*: $(T_k - q)(T_k + q^{-1}) = 0$, equivalently $T_k^2 = (q - q^{-1})\, T_k + 1$;

(ii) *Braid relations*: $T_kT_{k+1}T_k = T_{k+1}T_kT_{k+1}$;

(iii) *Far commutativity*: $T_kT_l = T_lT_k$ for $|k-l| > 1$.

At $q = 1$, the quadratic relation becomes $T_k^2 = 1$, recovering $\mathbb{C}[S_n]$ with $T_k \mapsto s_k$.

Remark 9.5. The eigenvalues of T_k are q and $-q^{-1}$ (from the quadratic relation). The dimension of $\mathrm{H}_n(q)$ is $n!$, the same as $\mathbb{C}[S_n]$, and $\mathrm{H}_n(q) \cong \mathbb{C}[S_n]$ as vector spaces (but not as algebras for $q \neq 1$).

9.3 The R-matrix action

The Hecke algebra acts on $V^{\otimes n}$ via the *R-matrix*, which replaces the simple transposition of tensor factors.

Definition 9.6. The *R-matrix action* of T_k on $V^{\otimes n}$ acts on positions k and $k{+}1$ by:

$$T_k \cdot (\cdots \otimes e_a \otimes e_b \otimes \cdots) = \begin{cases} q\,(\cdots \otimes e_a \otimes e_b \otimes \cdots) & \text{if } a = b, \\ (\cdots \otimes e_b \otimes e_a \otimes \cdots) & \text{if } a < b, \\ (\cdots \otimes e_b \otimes e_a \otimes \cdots) + (q - q^{-1})\,(\cdots \otimes e_a \otimes e_b \otimes \cdots) & \text{if } a > b, \end{cases}$$

where only positions k and $k{+}1$ are affected.

At $q = 1$ this reduces to the transposition s_k (swapping positions k and $k{+}1$). For $q \neq 1$, T_k still "mostly swaps" but adds a correction $(q - q^{-1})$ when the larger index is on the left.

9.4 Commutativity

Theorem 9.7 (Jimbo [4])**.** *The actions of $U_q(\mathfrak{gl}_N)$ and $\mathrm{H}_n(q)$ on $V^{\otimes n}$ commute: for all $x \in U_q(\mathfrak{gl}_N)$ and $h \in \mathrm{H}_n(q)$,*

$$x \cdot (h \cdot v) = h \cdot (x \cdot v) \qquad \text{for all } v \in V^{\otimes n}.$$

Remark 9.8. Unlike the classical case (Proposition 1.1), this is *not* obvious. The classical proof relied on GL_N acting within each factor and S_n permuting factors—these two operations are manifestly independent. In the quantum case, the coproduct makes $U_q(\mathfrak{gl}_N)$ act asymmetrically across factors, and the R-matrix makes $\mathrm{H}_n(q)$ act non-trivially on values within each factor. That they still commute is a consequence of the *Yang–Baxter equation* satisfied by the R-matrix.

10 The decomposition (q-deformed)

10.1 Partitions and tableaux (recap)

The combinatorial objects—partitions, Young diagrams, standard Young tableaux (SYT), and semistandard Young tableaux (SSYT)—are identical to the classical case (Section 2). We use the same notation: $\lambda \vdash n$ for partitions, $\mathrm{SYT}(\lambda)$ for standard tableaux, $\mathrm{SSYT}(\lambda, N)$ for semistandard tableaux.

10.2 The q-Schur–Weyl decomposition

Theorem 10.1 (q-Schur–Weyl duality)**.** *As a $(U_q(\mathfrak{gl}_N),\, \mathrm{H}_n(q))$-bimodule,*

$$V^{\otimes n} \;\cong \bigoplus_{\substack{\lambda \vdash n \\ \ell(\lambda) \leqslant N}} W_q^\lambda \otimes S_q^\lambda, \tag{14}$$

where:

- *W_q^λ is the irreducible $U_q(\mathfrak{gl}_N)$-module of highest weight λ,*
- *S_q^λ is the irreducible $\mathrm{H}_n(q)$-module indexed by λ.*

Moreover:

(a) The natural map $\mathrm{H}_n(q) \to \mathrm{End}_{U_q(\mathfrak{gl}_N)}(V^{\otimes n})$ is surjective (and injective when $n \leqslant N$).

(b) The natural map $U_q(\mathfrak{gl}_N) \to \mathrm{End}_{\mathrm{H}_n(q)}(V^{\otimes n})$ is surjective.

Remark 10.2. The dimensions are unchanged: $\dim W_q^\lambda = \dim W^\lambda = |\,\mathrm{SSYT}(\lambda, N)|$ and $\dim S_q^\lambda = \dim S^\lambda = |\,\mathrm{SYT}(\lambda)|$. The deformation changes the algebra structure but not the combinatorics.

11 Explicit matrix elements for $\mathrm{H}_n(q)$: q-Young's seminormal form

11.1 Content and axial distance (recap)

The *content* of a box in row r, column c of a Young diagram is $\mathrm{cont}(r,c) = c - r$ (Definition 3.1). For a standard Young tableau T, $\mathrm{cont}_T(k)$ is the content of the box containing k. The *axial distance* is $\rho_T(k) = \mathrm{cont}_T(k{+}1) - \mathrm{cont}_T(k)$ (unchanged from the classical case).

11.2 The q-seminormal representation

Definition 11.1. For each $\lambda \vdash n$, the *q-seminormal representation* is the representation of $\mathrm{H}_n(q)$ on the vector space $\widetilde{S}_q^\lambda = \bigoplus_{T \in \mathrm{SYT}(\lambda)} \mathbb{C} \cdot v_T$ defined by:

$$T_k \cdot v_T = \frac{q - q^{-1}}{1 - q^{-2\rho}}\, v_T + \frac{1 - q^{-2\rho-2}}{1 - q^{-2\rho}}\, v_{s_k \cdot T}, \tag{15}$$

where $\rho = \rho_T(k)$ and $s_k \cdot T$ is the tableau with entries k and $k{+}1$ swapped.

If $s_k \cdot T$ is not a valid standard tableau ($|\rho| = 1$), the second term is absent and $T_k \cdot v_T = q\, v_T$ (when $\rho = 1$) or $T_k \cdot v_T = -q^{-1}\, v_T$ (when $\rho = -1$).

Remark 11.2 (Degeneration to $q = 1$). As $q \to 1$:

- the diagonal coefficient $\dfrac{q - q^{-1}}{1 - q^{-2\rho}} \to \dfrac{1}{\rho}$ (by L'Hôpital),
- the off-diagonal coefficient $\dfrac{1 - q^{-2\rho-2}}{1 - q^{-2\rho}} \to \dfrac{\rho + 1}{\rho} = 1 + \dfrac{1}{\rho}$,

recovering the classical seminormal form (Remark 3.5).

Remark 11.3. The representing matrix for T_k in the 2×2 block $\{v_T, v_{s_k \cdot T}\}$ is *not* symmetric: the $(2,1)$-entry uses $\rho' = -\rho$, giving a different off-diagonal coefficient. This is analogous to the classical seminormal form and contrasts with the orthogonal form of Part I where the matrix was symmetric.

Example 11.4. For $\lambda = (2,1) \vdash 3$ with standard tableaux $T_1 = \begin{array}{|c|c|}\hline 1 & 2 \\ \hline 3 \\ \cline{1-1}\end{array}$ and $T_2 = \begin{array}{|c|c|}\hline 1 & 3 \\ \hline 2 \\ \cline{1-1}\end{array}$.

The axial distances are $\rho_{T_1}(1) = 1$, $\rho_{T_1}(2) = -2$, $\rho_{T_2}(1) = -1$, $\rho_{T_2}(2) = 2$.

Since $|\rho_{T_1}(1)| = |\rho_{T_2}(1)| = 1$:

$$T_1 \cdot v_{T_1} = q\, v_{T_1}, \qquad T_1 \cdot v_{T_2} = -q^{-1}\, v_{T_2}.$$

For T_2 (with $\rho = \rho_{T_1}(2) = -2$):

$$T_2 = \begin{pmatrix} \dfrac{q - q^{-1}}{1 - q^4} & \dfrac{1 - q^{4-2}}{1 - q^4} \\[2ex] \dfrac{1 - q^{-4+2}}{1 - q^{-4}} & \dfrac{q - q^{-1}}{1 - q^{-4}} \end{pmatrix} = \begin{pmatrix} -\dfrac{q - q^{-1}}{q^4 - 1} & \dfrac{1 - q^2}{1 - q^4} \\[2ex] \dfrac{1 - q^{-2}}{1 - q^{-4}} & \dfrac{q - q^{-1}}{1 - q^{-4}} \end{pmatrix}.$$

One can verify $(T_1 - q)(T_1 + q^{-1}) = 0$, $(T_2 - q)(T_2 + q^{-1}) = 0$, and $T_1 T_2 T_1 = T_2 T_1 T_2$.

12 Explicit matrix elements for $U_q(\mathfrak{gl}_N)$: the q-GT basis

12.1 q-GT patterns (recap)

Gel'fand–Tsetlin patterns are unchanged: a GT pattern Λ of shape λ is a triangular array satisfying the interlacing conditions $\lambda_{k,i} \geqslant \lambda_{k-1,i} \geqslant \lambda_{k,i+1}$ (Definition 4.1). The bijection between GT patterns and semistandard tableaux is the same.

12.2 The $U_q(\mathfrak{gl}_N)$ action in the q-GT basis

Write v_Λ for the basis vector of W_q^λ indexed by Λ. In the shifted coordinates $\ell_{k,i} = \lambda_{k,i} - i$ (Remark 4.4), the action of $U_q(\mathfrak{gl}_N)$ is:

Proposition 12.1 (q-GT formulae)**.**

$$K_k\, v_\Lambda = q^{\sum_{i=1}^{k} \lambda_{k,i} - \sum_{i=1}^{k-1} \lambda_{k-1,i}}\, v_\Lambda, \tag{16}$$

$$E_k\, v_\Lambda = -\sum_{i=1}^{k} \frac{\displaystyle\prod_{j=1}^{k+1} [\ell_{k,i} - \ell_{k+1,j}]_q \prod_{j=1}^{k-1} [\ell_{k,i} - \ell_{k-1,j}]_q}{\displaystyle\prod_{\substack{j=1 \\ j\neq i}}^{k} [\ell_{k,i} - \ell_{k,j}]_q}\, v_{\Lambda + \delta_{k,i}}, \tag{17}$$

$$F_k\, v_\Lambda = \sum_{i=1}^{k} \frac{\displaystyle\prod_{j=1}^{k+1} [\ell_{k,i} - \ell_{k+1,j}]_q \prod_{j=1}^{k-1} [\ell_{k,i} - \ell_{k-1,j}]_q}{\displaystyle\prod_{\substack{j=1 \\ j\neq i}}^{k} [\ell_{k,i} - \ell_{k,j}]_q}\, v_{\Lambda - \delta_{k,i}}, \tag{18}$$

where $[x]_q = (q^x - q^{-x})/(q - q^{-1})$ *and the term is zero if the result is not a valid GT pattern.*

Remark 12.2. Comparing with the classical GT formulae ((9)–(10)), the only change is that each linear factor $(\ell_{k,i} - \ell_{k',j})$ is replaced by its q-analogue $[\ell_{k,i} - \ell_{k',j}]_q$. The structure—products over adjacent rows in the numerator, Vandermonde product within the same row in the denominator—is identical.

The Cartan action (16) exponentiates the classical weight: K_k has eigenvalue q^{w_k} where w_k is the classical $E_{k,k}$ eigenvalue (6).

13 The RSK correspondence (q-deformed setting)

The RSK correspondence is purely combinatorial and does not involve the parameter q. The bijection

$$\mathrm{RSK} : \{1, \ldots, N\}^n \xrightarrow{\sim} \bigsqcup_{\substack{\lambda \vdash n \\ \ell(\lambda) \leqslant N}} \mathrm{SSYT}(\lambda, N) \times \mathrm{SYT}(\lambda)$$

is identical to Definition 5.1. The insertion algorithm, recording tableau, and all combinatorial properties (Proposition 5.3) are unchanged.

14 RSK is the q-Schur–Weyl decomposition

The interpretive result of Section 6 carries over verbatim to the q-deformed setting.

Theorem 14.1 (q-RSK as Schur–Weyl)**.** *Under the isomorphism* (14)*, the basis element* $e_{\mathbf{i}}$ *corresponds to*

$$e_{\mathbf{i}} \;\longleftrightarrow\; w_P \otimes v_Q \;\in\; W_q^\lambda \otimes S_q^\lambda,$$

where $(P, Q) = \mathrm{RSK}(\mathbf{i})$*. The insertion tableau* P *indexes the* $U_q(\mathfrak{gl}_N)$*-irrep basis (via the* q*-GT identification), and the recording tableau* Q *indexes the* $\mathrm{H}_n(q)$*-irrep basis (via the SYT basis of the* q*-seminormal representation).*

Remark 14.2. The table of Section 6.2 applies without change: P is sensitive to what letters appear ($U_q(\mathfrak{gl}_N)$ symmetry), Q is sensitive to the order of arrival ($\mathrm{H}_n(q)$ symmetry), and the shape λ determines which summand.

15 The bridge: q-Jucys–Murphy elements

15.1 Motivation

The strategy of Part I carries over: use the chain $\mathrm{H}_1(q) \subset \mathrm{H}_2(q) \subset \cdots \subset \mathrm{H}_n(q)$ to build a maximal commutative subalgebra whose spectral projectors carve out the irreducible summands.

In the classical case, $J_k = X_k - X_{k-1}$ was an additive "peeling off" of the new element. In the q-deformed case, the construction becomes multiplicative.

15.2 The q-Jucys–Murphy elements

Definition 15.1. The *q-Jucys–Murphy elements* of $\mathrm{H}_n(q)$ are

$$L_1 = 1, \qquad L_k = T_{k-1}\, L_{k-1}\, T_{k-1}, \quad k = 2, \ldots, n.$$

Expanding the recursion: $L_2 = T_1^2 = (q - q^{-1})T_1 + 1$, $L_3 = T_2 L_2 T_2 = T_2 T_1^2 T_2$, and so on. Each L_k is a product of Hecke generators involving $T_1, \ldots, T_{k-1}$.

Remark 15.2 (Degeneration to $q = 1$). At $q = 1$, the recursion gives $L_k = s_{k-1} L_{k-1} s_{k-1}$. From the classical recursion $J_k = s_{k-1} J_{k-1} s_{k-1} + s_{k-1}$ (Lemma 7.3), we see that L_k and J_k are related but not identical at $q = 1$: the q-JM elements are *multiplicative* while the classical ones are *additive*. The precise relation is $L_k|_{q=1} = J_k + 1$ (the identity shift absorbs the $+s_{k-1}$ correction).

15.3 Commutativity

Lemma 15.3. *The elements* $L_1, \ldots, L_n$ *commute pairwise.*

Proof. We first show that L_k commutes with T_i for $i \neq k-1, k$.

For $i < k-1$: by induction on k. The base case $L_2 = (q-q^{-1})T_1+1$ trivially commutes with T_i for $i > 2$ (far commutativity). For the inductive step, $L_k = T_{k-1}L_{k-1}T_{k-1}$, and T_i commutes with T_{k-1} (since $i < k-1$ implies $|i-(k-1)| > 1$) and with L_{k-1} (by induction), hence with L_k.

For $i > k$: T_i commutes with all of $T_1, \ldots, T_{k-1}$, hence with L_k.

Now L_k and L_l commute for $|k-l| \geqslant 2$: $L_l = T_{l-1}L_{l-1}T_{l-1}$ involves only T_{l-1} and L_{l-1}, both of which commute with L_k (the former because $l-1 > k$, the latter by induction).

For L_k and L_{k+1}: $L_{k+1} = T_kL_kT_k$, so $L_kL_{k+1} = L_kT_kL_kT_k$ and $L_{k+1}L_k = T_kL_kT_kL_k$. These are equal because $L_kT_kL_k = T_kL_kT_k \cdot L_k \cdot T_k^{-1}$... more precisely, one verifies directly from the Hecke relations and the recursion that $T_kL_kT_k \cdot L_k = L_k \cdot T_kL_kT_k$. □

The L_k generate the *q-Gelfand–Tsetlin subalgebra* of $\mathrm{H}_n(q)$. By "commutative subalgebra" we mean that the L_k commute with each other—they do not commute with arbitrary elements of $\mathrm{H}_n(q)$.

15.4 The eigenvalue theorem

Proposition 15.4. *In the q-seminormal representation $\widetilde{S}^\lambda_q$ (Definition 11.1), the q-Jucys–Murphy element L_k acts diagonally in the SYT basis:*

$$L_k \cdot v_T = q^{2\,\mathrm{cont}_T(k)}\, v_T,$$

where $\mathrm{cont}_T(k) = c - r$ is the content of the box in row r, column c that contains entry k in T.

Proof. By induction on k, using the recursion $L_k = T_{k-1}L_{k-1}T_{k-1}$.

Base case ($k = 1$): $L_1 = 1$, and $q^{2\cdot 0} = 1$ (entry 1 is always in box $(1,1)$ with content 0).

Inductive step: Assume $L_{k-1}\, v_T = q^{2c_1}\, v_T$ for all T, where $c_1 = \mathrm{cont}_T(k-1)$. Set $c_2 = \mathrm{cont}_T(k)$, $\rho = c_2 - c_1$, and let $T' = s_{k-1}\cdot T$. We must show $L_k\, v_T = T_{k-1}L_{k-1}T_{k-1}\, v_T = q^{2c_2}\, v_T$.

Write $\alpha = (q-q^{-1})/(1-q^{-2\rho})$ and $\beta = (1-q^{-2\rho-2})/(1-q^{-2\rho})$ for the diagonal and off-diagonal coefficients in (15), and $\alpha' = (q-q^{-1})/(1-q^{2\rho})$, $\beta' = (1-q^{2\rho-2})/(1-q^{2\rho})$ for the corresponding coefficients for T' (where $\rho_{T'}(k-1) = -\rho$). Compute step by step:

$$\begin{aligned} T_{k-1}\, v_T &= \alpha\, v_T + \beta\, v_{T'}, \\ L_{k-1}(\alpha\, v_T + \beta\, v_{T'}) &= \alpha\, q^{2c_1}\, v_T + \beta\, q^{2c_2}\, v_{T'} \quad \text{(by induction, since } \mathrm{cont}_{T'}(k-1) = c_2\text{)}, \\ T_{k-1}(\alpha\, q^{2c_1}\, v_T + \beta\, q^{2c_2}\, v_{T'}) &= \alpha q^{2c_1}(\alpha\, v_T + \beta\, v_{T'}) + \beta q^{2c_2}(\beta'\, v_T + \alpha'\, v_{T'}). \end{aligned}$$

The v_T coefficient is $\alpha^2 q^{2c_1} + \beta\beta' q^{2c_2}$. One verifies (using $q^{2c_2} = q^{2\rho}q^{2c_1}$ and the explicit formulae for α, β, β') that this equals q^{2c_2}.

The $v_{T'}$ coefficient is $\alpha\beta q^{2c_1} + \beta\alpha' q^{2c_2} = \beta(\alpha q^{2c_1} + \alpha' q^{2c_2})$. Since $\alpha + \alpha' = q - q^{-1}$ and $\alpha q^{2c_1} + \alpha' q^{2c_2} = \alpha q^{2c_1} + (q - q^{-1} - \alpha)q^{2\rho}q^{2c_1} = q^{2c_1}[\alpha(1-q^{2\rho}) + (q-q^{-1})q^{2\rho}]$, and substituting $\alpha(1-q^{2\rho}) = -(q-q^{-1})q^{2\rho}$ (from the definition of α), this is 0.

So $L_k\, v_T = q^{2c_2}\, v_T = q^{2\,\mathrm{cont}_T(k)}\, v_T$. □

Remark 15.5. At $q = 1$: $L_k|_{q=1} = 1$ for all k (since $L_k = T_{k-1}L_{k-1}T_{k-1}$ and $T_k|_{q=1} = s_k$ with $s_k^2 = 1$, giving $L_k = s_{k-1}L_{k-1}s_{k-1}$ with $L_1 = 1$, so $L_k = 1$ for all k). This is consistent with $q^{2\,\mathrm{cont}_T(k)}|_{q=1} = 1$. The classical JM elements J_k (with non-trivial eigenvalues $\mathrm{cont}_T(k)$) are recovered via the infinitesimal version: $J_k = \lim_{q\to 1} \frac{L_k - 1}{q^2-1}$.

15.5 Content separation and maximality

Lemma 7.6 (different SYTs have different content vectors) applies unchanged, since content vectors do not depend on q. The q-eigenvalues $q^{2\,\mathrm{cont}_T(k)}$ are distinct whenever the contents are distinct (since q is not a root of unity), so the q-GZ subalgebra separates all basis vectors.

Theorem 15.6. *The q-GZ subalgebra generated by $L_1, \ldots, L_n$ is a maximal commutative subalgebra of $\mathrm{H}_n(q)$.*

Proof. The same argument as Theorem 7.7: by Artin–Wedderburn, $\mathrm{H}_n(q) \cong \bigoplus_\lambda \mathrm{Mat}_{d_\lambda}(\mathbb{C})$ (since q is not a root of unity), the q-GZ subalgebra acts diagonally in the SYT basis with distinct eigenvalues, hence contains all diagonal matrices in each block. □

15.6 q-spectral projectors

Definition 15.7. For each $Q \in \mathrm{SYT}(\lambda)$, the *$q$-primitive idempotent* is

$$e_Q^{(q)} \;=\; \prod_{k=2}^{n} \prod_{\substack{c \neq q^{2\,\mathrm{cont}_Q(k)} \\ c \text{ an eigenvalue of } L_k}} \frac{L_k - c}{q^{2\,\mathrm{cont}_Q(k)} - c} \;\in\; \mathrm{H}_n(q).$$

Since $e_Q^{(q)} \in \mathrm{H}_n(q)$ and the $U_q(\mathfrak{gl}_N)$-action commutes with $\mathrm{H}_n(q)$ (Theorem 9.7), the subspace $e_Q^{(q)} \cdot V^{\otimes n}$ is an irreducible $U_q(\mathfrak{gl}_N)$-module isomorphic to W_q^λ, exactly as in the classical case (Proposition 7.10).

16 Degeneration: from q to 1

	Classical ($q = 1$)	**q-deformed**
Symmetry algebra	$\mathbb{C}[S_n]$	$\mathrm{H}_n(q)$
Generators	s_k with $s_k^2 = 1$	T_k with $(T_k - q)(T_k + q^{-1}) = 0$
Dual algebra	$U(\mathfrak{gl}_N)$	$U_q(\mathfrak{gl}_N)$
Action on $V^{\otimes n}$	diagonal + permutation	coproduct + R-matrix
JM elements	$J_k = \sum_{i<k}(i\ k)$ (additive)	$L_k = T_{k-1}L_{k-1}T_{k-1}$ (multiplicative)
JM recursion	$J_k = s_{k-1}J_{k-1}s_{k-1} + s_{k-1}$	$L_k = T_{k-1}L_{k-1}T_{k-1}$
JM eigenvalues	$\mathrm{cont}_T(k) \in \mathbb{Z}$	$q^{2\,\mathrm{cont}_T(k)} \in \mathbb{C}^*$
GT coefficients	$(\ell_{k,i} - \ell_{k',j})$	$[\ell_{k,i} - \ell_{k',j}]_q$
Cartan eigenvalue	$\sum \lambda_{k,i} - \sum \lambda_{k-1,i}$	$q^{\sum \lambda_{k,i} - \sum \lambda_{k-1,i}}$
Combinatorics	unchanged: SYT, SSYT, GT patterns, RSK	

The deformation is systematic: *integers become q-integers, contents become q-contents, linear factors become q-linear factors.* The combinatorial backbone—partitions, tableaux, RSK—is untouched. The algebraic structures deform continuously, and the representation theory deforms with them.

Remark 16.1. The q-JM recursion $L_k = T_{k-1}L_{k-1}T_{k-1}$ is *simpler* than the classical one $J_k = s_{k-1}J_{k-1}s_{k-1} + s_{k-1}$. The additive correction $+s_{k-1}$ in the classical case is an artifact of the degeneration $q \to 1$: the quadratic relation $(T_k - q)(T_k + q^{-1}) = 0$ absorbs this correction, and it only reappears when the quadratic degenerates to $s_k^2 = 1$.

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